

Iris Flower Classification

Project Overview

This project was developed as part of the CodeAlpha Data Science Internship. The main objective of this project is to classify Iris flowers into different species using Machine Learning techniques based on flower measurements.

The flower species used in this project are:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Objective

The goal of this project is to:

- Understand basic classification concepts in Machine Learning
- Analyze the Iris flower dataset
- Train a Machine Learning model
- Evaluate model accuracy and performance
- Predict flower species using flower measurements

Dataset Information

The dataset contains the following flower measurements:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable:

- Species

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Machine Learning Algorithm Used

K-Nearest Neighbors (KNN)

KNN is a supervised Machine Learning algorithm used for classification problems. It classifies data points based on similarity with nearby data points.

Project Workflow

1. Import Libraries

2. Load Dataset
3. Data Understanding
4. Data Cleaning
5. Exploratory Data Analysis
6. Data Visualization
7. Feature Selection
8. Train-Test Split
9. Model Building
10. Model Training
11. Prediction
12. Model Evaluation
13. Custom Prediction

Data Visualization

The following visualizations were created in this project:

- Count Plot
- Pair Plot
- Correlation Heatmap
- Confusion Matrix Heatmap

These visualizations helped in understanding relationships between features and species.

Model Performance

The KNN model achieved high accuracy on the test dataset.

Evaluation methods used:

- Accuracy Score
- Confusion Matrix
- Classification Report

Conclusion

This project successfully demonstrates the implementation of Machine Learning classification using the Iris flower dataset.

The K-Nearest Neighbors (KNN) algorithm performed very well and achieved high accuracy in predicting flower species.

Through this project, I learned:

- Data preprocessing
- Data visualization
- Classification algorithms
- Model training
- Model evaluation
- Prediction using Machine Learning

This project helped me improve my practical understanding of Data Science and Machine Learning concepts.